

1. Administrative & Study Identification

Full Study Title: A Phase Ib/II, Open-Label, Multicenter, Randomized Umbrella Study Evaluating The Efficacy And Safety Of Multiple Immunotherapy-Based Treatment Combinations In Patients With Metastatic Non-Small Cell Lung Cancer (Morpheus-Lung) **Brief Title:** A Study Of Multiple Immunotherapy-Based Treatment Combinations In Participants With Metastatic Non-Small Cell Lung Cancer (Morpheus- Non-Small Cell Lung Cancer) **Protocol Number:** B039610 (2017-001267-21) **NCT Number:** NCT03337698 **Posting ID:** 87245366133851017 **Sponsor/Company Name:** Hoffmann-La Roche / Roche **Investigational Product:** Multiple immunotherapy-based combinations (e.g., Atezolizumab, Bevacizumab, Tocilizumab, Tiragolumab, Gemcitabine [Gemzar, Infugem]) **Phase of Development:** Phase I, Phase II **Trial Status:** Active, not recruiting (Active but not currently recruiting) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the objective response rate (ORR) of multiple immunotherapy-based treatment combinations in patients with metastatic non-small cell lung cancer (NSCLC). **Secondary Objectives:** To evaluate progression-free survival (PFS), overall survival (OS), percentage of participants who are alive at Month 6 and Month 12, disease control rate (DCR), duration of response (DOR), pharmacokinetics (PK), and the safety/tolerability of the experimental combinations.

2.2 Background & Rationale While cancer immunotherapy (CIT) has demonstrated significant survival benefits in multiple cancers, only subsets of patients experience durable responses with CIT monotherapy. Thus, CIT combinations may be needed to address the mechanisms that allow cancers to escape anti-tumor immunity. The MORPHEUS platform consists of multiple open-label, randomized, umbrella trials designed to identify effective CIT combinations rapidly and flexibly.

Two cohorts will be enrolled in parallel: Cohort 1 consists of participants with high tumor PD-L1 expression receiving no prior systemic therapy, and Cohort 2 consists of participants progressing on a platinum-containing regimen and a PD-L1/PD-1 checkpoint inhibitor. Participants experiencing disease progression, loss of clinical benefit, or unacceptable toxicity during Stage 1 may be eligible to continue treatment with a different treatment regimen (Stage 2).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Study)
Control Type: Active Comparator **Blinding:** Open-label
Randomization: Randomized

3.2 Planned Enrollment & Duration Target N: 314 participants
Number of Sites: ~37 global locations **Estimated Study Start:** 02-Jan-2018 **Estimated Primary Completion:** TBD / Active, not recruiting

4. Subject Selection (Eligibility Criteria)

4.1 General Inclusion Criteria 1. Eastern Cooperative Oncology Group (ECOG) performance Status of 0 or 1. 2. Histologically or cytologically confirmed metastatic, non-squamous or squamous Non-Small Cell Lung Cancer (NSCLC). 3. Measurable disease (at least one target lesion). 4. Life expectancy greater than or equal to 3 months. 5. Adequate hematologic and end-organ function. 6. Tumor accessible for biopsy. 7. For women of childbearing potential: agreement to remain abstinent or use contraceptive measures and refrain from donating eggs. 8. For men: agreement to remain abstinent or use contraceptive measures, and refrain from donating sperm.

Cohort 1 Inclusion: No prior systemic therapy for metastatic NSCLC; high tumor PD-L1 expression (Tumor Proportion Score [TPS] or TCs $\geq$ 50% or TC3). **Cohort 2 Inclusion:** Disease progression during or following treatment for metastatic or locally advanced, inoperable NSCLC (including a platinum-containing regimen and a PD-L1/PD-1 checkpoint inhibitor).

4.2 Exclusion Criteria 1. Prior allogeneic stem cell or solid organ transplantation. 2. Uncontrolled pleural effusion, pericardial effusion, or ascites requiring recurrent drainage procedures (once monthly or more frequently). 3. Symptomatic, untreated, or actively progressing central nervous system (CNS) metastases or history of leptomeningeal disease. 4. Active or history of autoimmune disease or immune deficiency. 5. History of idiopathic pulmonary fibrosis, organizing pneumonia (e.g., bronchiolitis obliterans), drug-induced pneumonitis, or idiopathic pneumonitis, or evidence of active pneumonitis on screening chest CT. 6. History of malignancy other than NSCLC within 2 years prior to screening. 7. Active tuberculosis. 8. Severe infection within 4 weeks prior to initiation of study treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Atezolizumab administered in combination with various agents (e.g., bevacizumab, tocilizumab, cobimetinib, gemcitabine [ATC: L01BC05], etc.) depending on the randomly assigned experimental arm in Stage 1. **Stage 2:** Participants who experience disease progression, loss of clinical benefit, or unacceptable toxicity during Stage 1 may cross over or continue treatment with a different treatment regimen. **Route of Administration:** Intravenous and/or Oral depending on the specific combination arm. **Duration of Treatment:** Until disease progression, loss of clinical benefit, or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures. **Prohibited:** Prior treatment with investigational therapy ≤ 28 days prior to study treatment initiation; other concurrent systemic anticancer therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy/Archival Tissue, Scans
Baseline	Day 0	Randomization, First Dose of combination therapy
Interim	Q3W / Q4W	Safety Assessments, Investigational Product Administration, Efficacy Scans, Pharmacokinetics
End of Study	Follow-up	Final Vital Signs, Exit Interview, Survival Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Percentage of Participants with Objective Response. **Measurement Method:** Assessed by the investigator according to Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1.

7.2 Secondary & Exploratory Endpoints

1. Progression Free Survival (PFS)
2. Overall Survival After Randomization
3. Percentage of Participants Who Are Alive at Month 6 and at Month 12
4. Duration of Response (DOR) and Disease Control Rate (DCR)
5. Pharmacokinetics (PK)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard safety evaluations throughout the trial, recording treatment-emergent AEs (TEAEs) graded by NCI CTCAE. **Serious Adverse Events (SAEs):** Reported within 24 hours to the Sponsor/Regulatory bodies. **Data Monitoring Committee (DMC):** Internal safety review committees continuously

monitor the safety across all umbrella arms to close those with unacceptable toxicity.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy-evaluable population (patients meeting eligibility and receiving treatment); Safety Set (all patients receiving at least one dose of study treatment). **Sample Size Justification:** Signal-seeking umbrella design with targeted enrollment of 314 participants allowing statistical comparison of multiple cohorts independently. **Handling of Missing Data:** Standard survival analysis methodologies (e.g., Kaplan-Meier estimation) and appropriate censoring rules applied for time-to-event endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Standard ongoing onsite and remote data monitoring ensuring protocol compliance and accurate reporting of RECIST criteria. **Audit Trail:** eCRF entry tracking, centralized radiological review, and stringent source data verification processes.

11. Study Identifiers & Governance

Primary ID: B039610 **Posting ID:** 87245366133851017 **Registry Number:** NCT03337698 **Sponsor/Lead Organization:** Hoffmann-La Roche / Roche **Study Title:** Morpheus-Lung **Official Regulatory Title:** A Phase Ib/II, Open-Label, Multicenter, Randomized Umbrella Study Evaluating The Efficacy And Safety Of Multiple Immunotherapy-Based Treatment Combinations In Patients With Metastatic Non-Small Cell Lung Cancer (Morpheus-Lung)

12. Clinical Framework (NSCLC Specific)

Disease Area / Primary Condition: Carcinoma, Non-Small-Cell Lung / Non-Small Cell Lung Carcinoma (Metastatic) **Diagnosis Threshold:** Histologically or cytologically confirmed metastatic disease, measurable by RECIST v1.1 **Medical Methodology:** Multiple Immunotherapy-Based Treatment Combinations (Umbrella Platform) **Optimization Target:** Improvement in Objective Response Rate (ORR) and overcoming immune-evasion mechanisms

13. Study Procedures & Methodology

Management Pathway: Step-wise assignment into Stage 1 treatment cohorts based on previous exposure to PD-1/PD-L1 inhibitors and platinum-based chemotherapy, with potential progression to Stage 2 crossover regimens. **Education Protocol (HCP):** Investigator site training on RECIST 1.1 evaluation, immune-related AE (irAE) management, and protocol compliance. **Education Protocol (Patient):** Informed consent detailing potential risks of novel combination therapies, including contraceptive requirements. **Quality Audit Mechanism:** Safety Review Committee tracking tumor responses and triggering the closure of ineffective or toxic arms dynamically.

14. Observation & Follow-Up Schedule

Total Duration: Until disease progression or loss of clinical benefit. **Onsite Visit Cadence:** Based on standard treatment cycles (e.g., Q3W or Q4W for immunotherapy infusions). **Remote Monitoring Frequency:** Intermittent survival follow-up contacts after treatment discontinuation. **Checklist Review Interval:** Regular tumor imaging every 6-9 weeks as per standard oncology trial protocols.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				